

Distribution of genetic paternity in primate groups

Supplementary Materials

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1. Bayesian model diagnostics

Table S1 summarizes convergence diagnostics for the Bayesian models reported in the main text. Q1 refers to the models used to answer Question 1 in the results in the main text; Q2/A1 and Q2/A2 refer to Question 2, Analysis 1 and 2 respectively, and Q3 refers to Question 3. Figures S1–S5 show the posterior predictive checks for each of these models. In each plot, the dark blue line represents the observed data distribution, and the lighter blue lines represent distributions simulated from the fitted model’s posterior. Significant overlap between the observed and simulated distributions indicates that the model captured the important features of the data.

Table S1. Bayesian model convergence diagnostics

Model	Max Rhat	Min Bulk ESS	Min Tail ESS	Divergent transitions
Q1: Phylogeny (Intercept-only, max)	1.0011	3414	5076	0
Q1: Phylogeny (Intercept-only, min)	1.0029	2341	3157	0
Q2/A1: Group composition	1.0003	3915	4164	0
Q2/A2: Competition/seasonality	1.0004	4176	4588	0
Q3: Paternity loss source	1.0002	4187	4666	0

Posterior Predictive Checks

Figure S1. Q1 Phylogeny (Intercept-only, max)

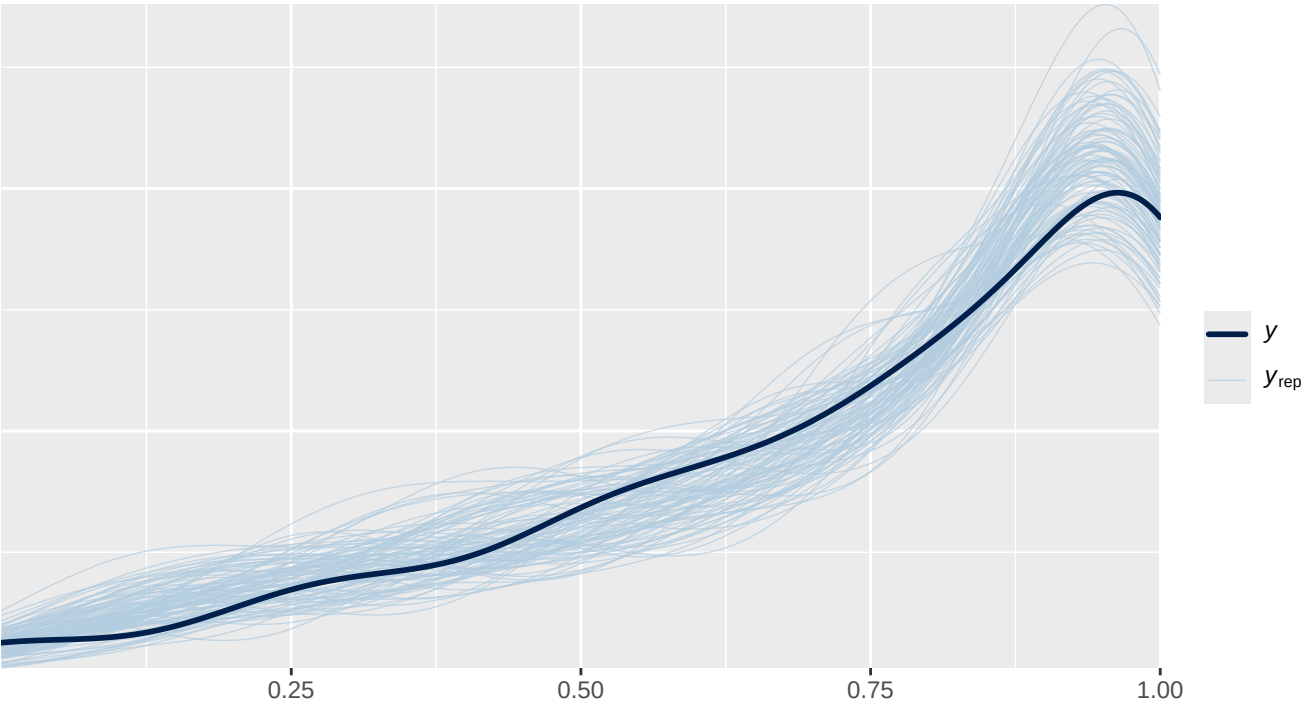


Figure S2. Q1 Phylogeny (Intercept-only, min)

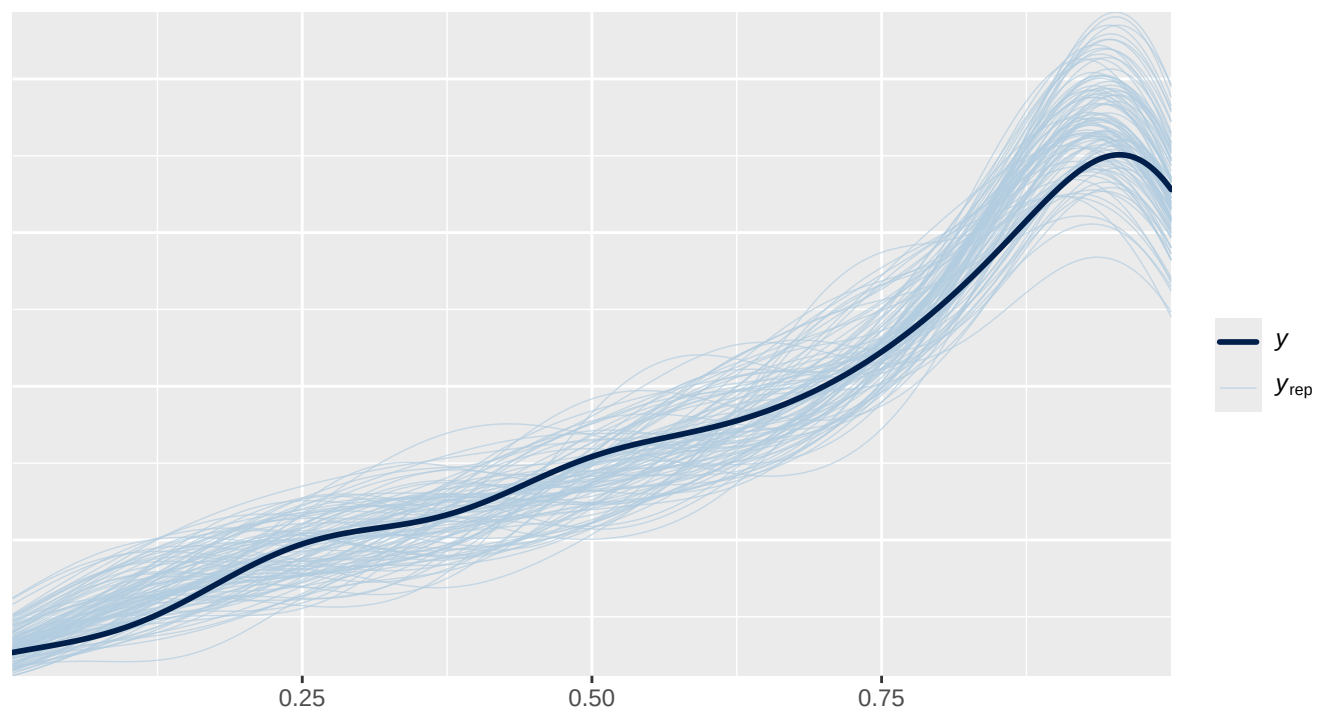


Figure S3. Q2/A1 Group composition

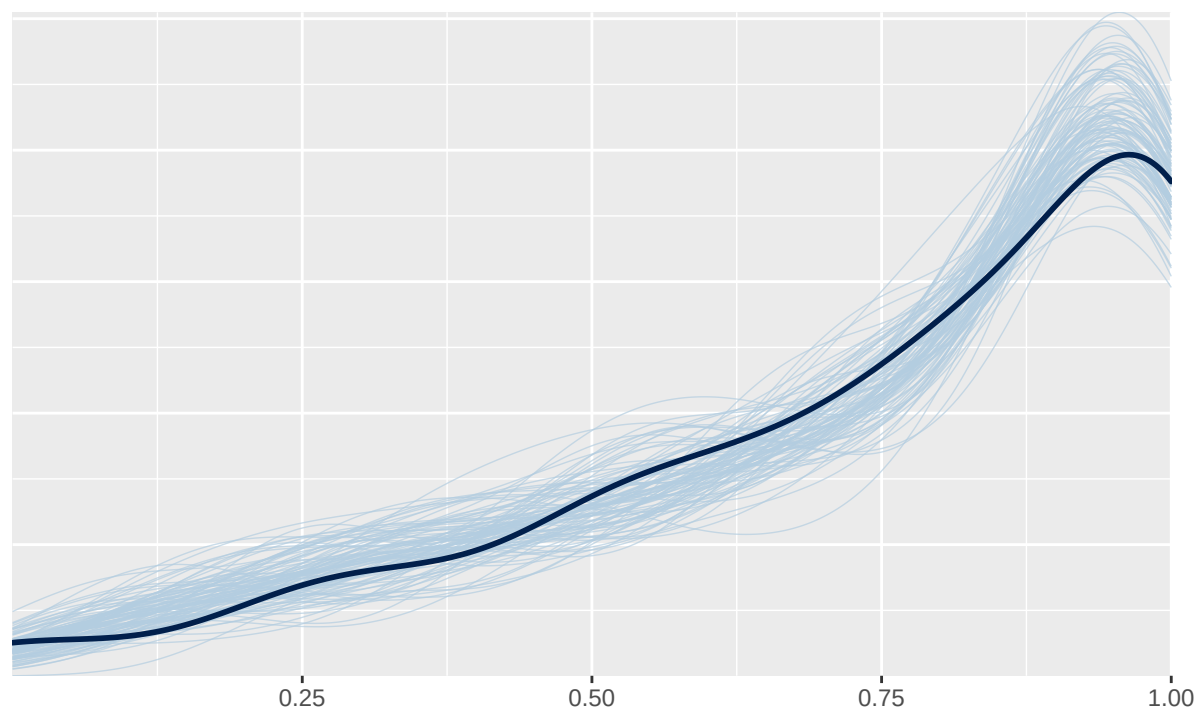


Figure S4. Q2/A2 Competition/seasonality

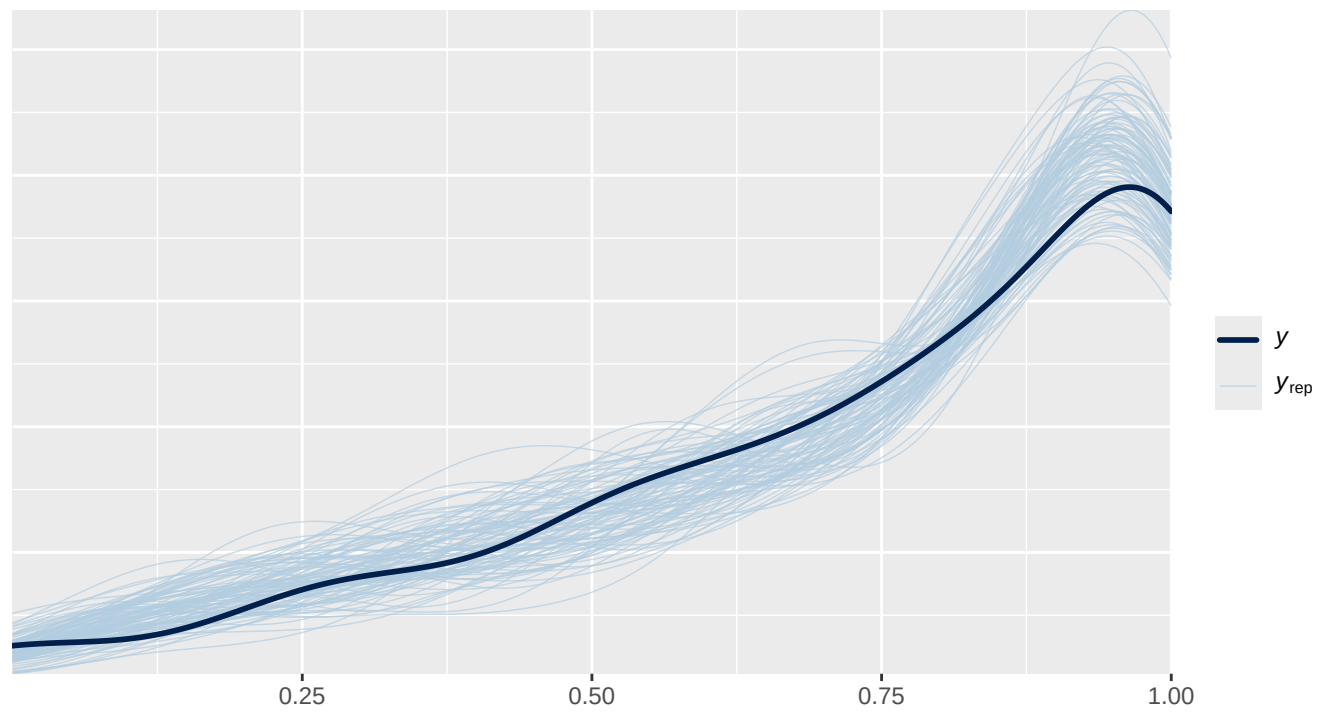
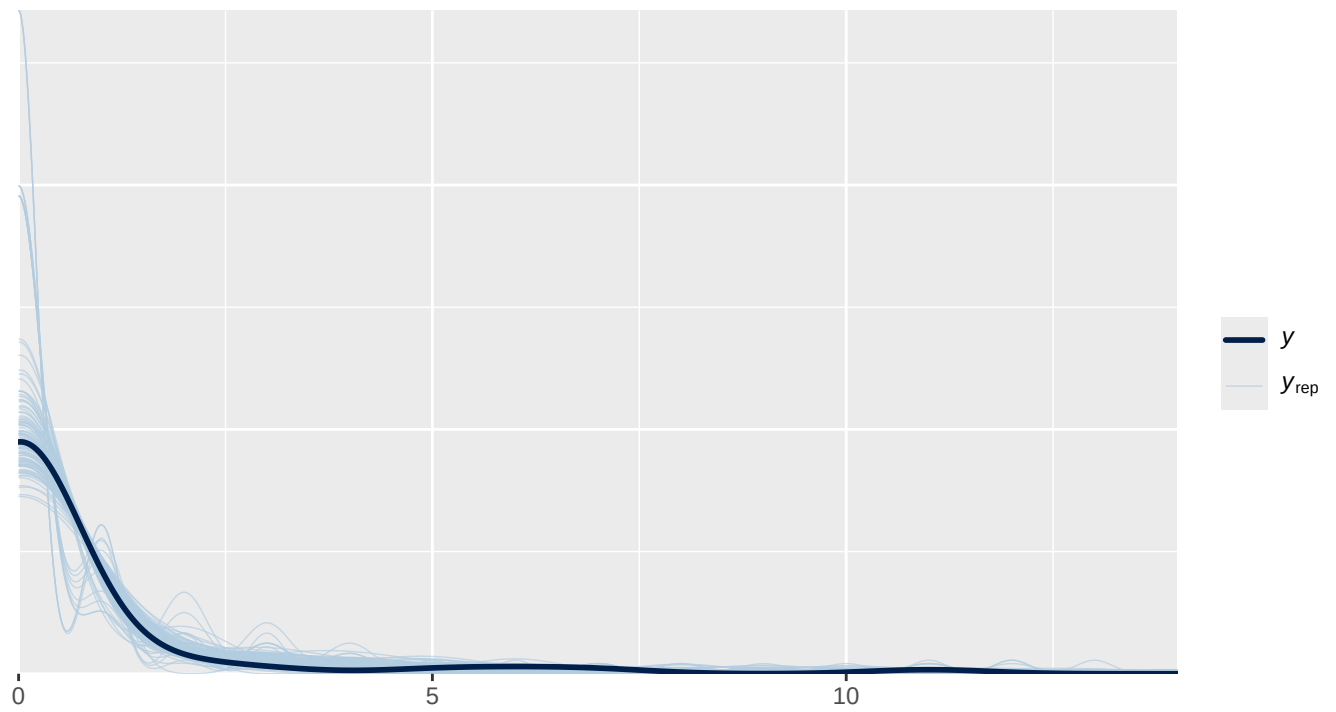


Figure S5. Q3 Paternity loss source



2. Sensitivity analyses: assignment of uncertain paternities

A. The effects of group composition (Question 2, Analysis 1 in main text)

In the main text the models for Question 2 use the maximum possible estimate of primary male paternity share; i.e., all uncertain paternities are assigned to primary males. For comparison purposes, here we present results from those models alongside results from models using minimum estimates, in which none of the uncertain paternities are assigned to primary males.

Table S2. Model results: Maximum primary male paternity share (in main text)

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept (Multi-male)	0.405	0.476	-0.622	1.286	1.000	17,269	13,670
Pair-living	0.482	0.300	-0.096	1.074	1.000	26,371	15,790
Single-male/multi-female	0.964	0.172	0.628	1.307	1.000	29,322	14,819

Table S3. Model results: minimum primary male paternity share

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept (Multi-male)	0.244	0.434	-0.723	1.032	1.001	12,258	10,985
Pair-living	0.705	0.291	0.136	1.277	1.000	19,555	15,331
Single-male/multi-female	1.333	0.165	1.010	1.660	1.000	23,841	15,093

Table S4. Model-derived predicted paternity share using maximum estimate (in main text)

Group type	Mean	95% CI
Multi-male	59.6%	34.9% - 78.3%
Pair-living	69.9%	46.2% - 85.6%
Single-male/multi-female	78.6%	57.9% - 90.8%

Table S5. Model-derived predicted paternity share using minimum estimate

Group type	Mean	95% CI
Multi-male	55.9%	32.7% - 73.7%
Pair-living	71.3%	49.2% - 85.4%
Single-male/multi-female	82.0%	63.9% - 91.6%

B. The effects of in-group competition and seasonality (Question 2, Analysis 2 in main text)

Table S6. Model results: Maximum primary male paternity share (in main text)

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.332	0.958	-2.214	1.542	1.000	23,627	15,987
Competition ¹	0.780	0.479	-0.152	1.702	1.000	21,751	14,993
Seasonality	0.649	0.824	-0.968	2.249	1.000	20,675	15,702
Seasonality ²	-0.162	0.214	-0.578	0.263	1.000	18,262	15,203
Competition*Seasonality	0.051	0.239	-0.415	0.524	1.000	21,846	14,552

¹Single-male versus multi-male group

Table S7. Model results: minimum primary male paternity share

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.208	0.953	-2.075	1.687	1.000	16,474	14,827
Competition ¹	0.908	0.458	0.014	1.809	1.000	15,966	14,648
Seasonality	0.493	0.809	-1.118	2.068	1.000	14,675	14,878
Seasonality ²	-0.177	0.204	-0.576	0.229	1.000	13,414	13,925
Competition*Seasonality	0.169	0.230	-0.281	0.617	1.000	15,660	14,594

¹Single-male versus multi-male group

Comparing the conditional effect of in-group competition between models Because the competition/seasonality models presented above contain interaction terms, the main effect of competition cannot be interpreted in isolation. In the below tables we compute the conditional effect of competition (i.e., the difference in predicted paternity share between single-male and multi-male groups) at different levels of seasonality for both the maximum and minimum paternity share models. The purpose of this is to determine whether the difference in the main effect coefficient on the competition parameter leads to a substantively different conclusion.

Table S8. Conditional effect of competition (maximum estimate)

Seasonality	Multi-male predicted	Single-male predicted	Difference	95% CI
Most seasonal	53.7%	71.6%	17.9%	6.2% to 30.3%
Moderately seasonal	57.6%	75.8%	18.2%	9.8% to 27.0%
Least seasonal	53.7%	73.0%	19.3%	6.1% to 33.9%

Table S9. Conditional effect of competition (minimum estimate)

Seasonality	Multi-male predicted	Single-male predicted	Difference	95% CI
Most seasonal	52.5%	75.3%	22.8%	10.9% to 34.8%
Moderately seasonal	51.7%	77.8%	26.1%	16.7% to 35.0%
Least seasonal	42.7%	73.1%	30.4%	16.1% to 44.0%

Interpretation In both models, competition has a meaningful effect on primary male paternity share: males in single-male groups are predicted to obtain a substantially higher share of paternities than males in multi-male groups at all levels of seasonality. However, the minimum estimate model produces larger point

estimates for this effect (~23–30% difference versus ~18–19% difference in the maximum estimate model). This is not surprising. Reassigning uncertain paternities away from primary males disproportionately reduces the predicted primary male paternity share in multi-male groups, where most of the ambiguity in paternity assignment occurs. However, there is considerable overlap in the 95% credible intervals between the two models at every level of seasonality. Thus, these two different methods of operationalizing primary male paternity share do not ultimately yield distinguishable estimates of the effect of competition. The qualitative conclusion (that competition is an important predictor of primary male paternity share, while seasonality is not) is robust to decisions about how to handle uncertain paternities.

C. Source of paternity losses in multi-male groups (Question 3 in main text)

In the main text we model the proportion of lost paternities attributable to extra-group males versus non-primary resident males using the minimum extra-group paternity (EGP) count (i.e., only confirmed extra-group paternities). Here we present that model alongside a model using the maximum possible EGP count, in which ambiguous paternities are assigned to extra-group males. Results are qualitatively similar regardless of which choice is made.

Table S10. Paternity loss source model: minimum EGP count (in main text)

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-1.294	1.060	-3.290	0.823	1.000	9,531	10,218

Table S11. Paternity loss source model: maximum EGP count

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-1.055	0.889	-2.620	0.818	1.000	6,798	12,667

Table S12. Model-derived predicted paternity loss source: minimum EGP count (in main text)

Loss source	Mean	95% CI
Extra-group males	25.7%	3.6% - 69.5%
Non-primary resident males	74.3%	30.5% - 96.4%

Table S13. Model-derived predicted paternity loss source: maximum EGP count

Loss source	Mean	95% CI
Extra-group males	28.6%	6.8% - 69.4%
Non-primary resident males	71.4%	30.6% - 93.2%

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